Chronos-Hydrodynamics: A Gross–Pitaevskii Superfluid Vacuum— Flat vs Threshold Turn-On in Casimir Ripple and Interferometric Visibility

George McNally
Independent Researcher
george@web-essentials.ie

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Abstract

We present **Chronos-Hydrodynamics (CH)**, a speculative framework in which space is a physical Bose–Einstein condensate obeying the Gross–Pitaevskii equation (GPE). Matter is a localized defect in the vacuum density field; gravity, time, and the effective metric emerge from hydrodynamic variables $(\rho, \mathbf{v}, S)$. A central design feature is **gradient gating**: in uniform, low-gradient vacuum the condensate is **observationally invisible**, explaining widespread null results in precision tests of empty space. Supersolid-linked observables turn on only when $|\nabla\rho|$ exceeds a critical scale $|\nabla\rho|_c$.

We derive dimensionally consistent relations including $G = \alpha_G c_s^2 \xi/\rho_{\text{in}}$ and $\beta_{\max} = \frac{3}{2}\xi^2/\hbar^2$, introduce a two-component density to address the cosmological-constant problem, and outline seven testable predictions. Analysis of real Fermi LAT data for GRB 090510 yields null quadratic dispersion ($p \approx 0.56$), consistent with gradient-gated CH on void paths but **not** distinguishable from standard quantum field theory (QFT) there.

The **primary CH-specific test** is Prediction #7: scan a laboratory knob that raises $|\nabla\rho|$ (Casimir gap d, sphere radius R) and compare **flat** (QFT) vs **threshold** (CH) models for Casimir ripple amplitude α_{eff} and interferometric visibility dip ΔV. We provide GPE boundary simulations, a pre-registered lab protocol, and open-source analysis code that returns confirm / rule-out / inconclusive verdicts from (k, O, σ) data.

Keywords: emergent gravity; Gross–Pitaevskii equation; superfluid vacuum; Casimir effect; gradient threshold; falsifiability

1 Introduction

1.1 What we believe (postulates)

CH is **not** established physics. It is a **research framework** with explicit postulates:

1. **Ontological:** Space is a complex order parameter $\psi(\mathbf{x}, t)$ in a macroscopic quantum state (BEC).
2. **Dynamical:** ψ obeys the GPE; Madelung variables (ρ, S) are fundamental hydrodynamic fields.
3. **Matter:** Mass is a **localized defect** (depletion) in ρ, not an object in passive space.
4. **Invisibility:** Uniform ρ with $|\nabla\rho| \approx 0$ yields **no observable structure**.

5. **Gradient gating:** Large $|\nabla\rho|$ drives supersolid order; observables scale with $\chi(|\nabla\rho|)$.

6. **Emergence:** G, c, and clock rates derive from $(\rho_{\text{in}}, \xi, c_s, m_{\text{grain}})$.

7. **Covariance:** Low-energy excitations see an **acoustic metric**; Lorentz symmetry is emergent in the comoving limit.

1.2 What we do not claim

- A complete Standard Model embedding or fermion derivation.
- That GRB nulls **prove** CH (they are also expected in QFT).
- That laptop simulations replace laboratory hardware.
- Numerological identities such as $m_{\text{vac}} = h$ (replaced by $m_{\text{grain}} = \hbar\omega_0/c^2$).

1.3 Scope of this manuscript

Most precision tests probe **uniform** vacuum and return **null**—which CH interprets as **by design**, not as falsification. The falsifiable claim is **gradient-correlated turn-on** (Prediction #7). This paper unifies theory, completed astrophysical null analysis, GPE knob maps, lab protocol, and analysis software for publication or collaboration outreach.

1.4 Motivation: the empty stage problem

General relativity treats spacetime as dynamic geometry; quantum field theory (QFT) treats the vacuum as the ground state of fields with enormous zero-point energy—naively $\sim 10^{120}$ times larger than the observed cosmological constant ρ_Λ. Neither picture offers a **mechanical** substrate whose local state could explain why empty space is so quiet in precision tests.

Chronos-Hydrodynamics (CH) proposes a fourth picture: space is a material **Bose–Einstein condensate (BEC)** whose Madelung fields $(\rho, \mathbf{v}, S)$ carry both hydrodynamics and an emergent acoustic metric [1, 2, 3]. Matter is not embedded in passive space but modeled as a **localized defect** (density depletion) in ρ. The key experimental design choice is **gradient gating**: in uniform vacuum ($|\nabla\rho| \approx 0$) the condensate is dynamically quiet and **observationally invisible**; supersolid-linked effects turn on only when $|\nabla\rho|$ exceeds $|\nabla\rho|_c$. That is why void-path GRB timing and smooth Casimir forces are **expected nulls**, not surprises—and why the discriminating test must **raise** $|\nabla\rho|$ in the laboratory.

1.5 Related work and what is new

CH builds on the Madelung hydrodynamic form of quantum mechanics [4], the Gross–Pitaevskii description of weakly interacting BECs [5, 6], and the **analog gravity** program in which phonons on a fluid see an emergent Lorentzian metric [1, 2]. Volovik's superfluid analogies to particle physics [3] motivate a material vacuum, but CH differs in four ways:

(i) **Gradient gating** — supersolid order is local ($\chi(|\nabla\rho|)$), not a global lattice visible everywhere.

(ii) **Two-component density** — only a fraction $\varepsilon \ll 1$ of inertial ρ_{in} gravitates at cosmological scales (Section 2).

(iii) **Defect ontology** — mass is depletion in ρ, not an external label in empty space.

(iv) **Pre-registered protocol** — Prediction #7 compares flat vs threshold $O(k)$ with a mandatory flat **control channel** (Section 6).

1.6 Contributions of this manuscript

For a reader who was not involved in our development process, this paper provides:

- Explicit postulates, limitations, and equations linking GPE fields to gated observables (Section 2–3).
- A completed reanalysis of real Fermi LAT extended data for GRB 090510, with honest interpretation of what null timing does and does **not** constrain (Section 4.2).
- GPE boundary estimates showing which laboratory knobs can approach $|\nabla\rho|_c$ (Section 5).
- A full Prediction #7 protocol: knob definitions, fit models, sample-size requirements, and automated verdict logic (Section 6; Appendix B).
- A lab-to-theory bridge: measured $k_c \to$ proposed ξ identifications and spherical defect gravity forecasts with self-consistent α_G calibration (Section 6.7).
- Open-source scripts that implement every analysis step cited here [7].

2 Theory: Gross–Pitaevskii vacuum

2.1 Fundamental equation

The vacuum order parameter $\psi(\mathbf{x}, t)$ satisfies

$$i\hbar\frac{\partial\psi}{\partial t} = \left(-\frac{\hbar^2}{2m_{\text{grain}}}\nabla^2 + V_{\text{ext}} + g|\psi|^2\right)\psi. \tag{1}$$

Madelung transformation:

$$\psi = \sqrt{\rho}\, e^{iS/\hbar}, \qquad \rho = |\psi|^2, \qquad \mathbf{v} = \frac{1}{\rho}\nabla S. \tag{2}$$

The **quantum potential** (central to the emergent-gravity sketch) is

$$Q = -\frac{\hbar^2}{2m_{\text{grain}}}\frac{\nabla^2\sqrt{\rho}}{\sqrt{\rho}}. \tag{3}$$

In the Madelung form, Q appears in the quantum Hamilton–Jacobi equation alongside classical pressure and external potentials. CH's Newtonian matching sketch (not derived fully here) identifies the gravitational potential with gradients of ρ and Q around localized defects: matter is a **displacement** of the vacuum density field, and G is a proportionality constant linking that displacement scale to observed free-fall (Eq. 6).

2.2 Grain scale and healing length

$$\omega_0 \equiv \frac{c_s}{\xi}, \qquad m_{\text{grain}} = \frac{\hbar\omega_0}{c^2} = \frac{\hbar c_s}{c^2\xi}, \tag{4}$$

$$c_s = \sqrt{\frac{g\rho_{\text{in}}}{m_{\text{grain}}}}, \qquad \xi = \frac{\hbar}{\sqrt{2m_{\text{grain}}g\rho_{\text{in}}}}. \tag{5}$$

2.3 Emergent Newton constant

$$\boxed{G = \frac{\alpha_G\, c_s^2\,\xi}{\rho_{\text{in}}}} \tag{6}$$

With measured G and a hypothesized ξ, this fixes ρ_{in} (or vice versa). α_G is a dimensionless coupling (often set to unity in sketches).

2.4 Two-component density

To address the cosmological-constant problem,

$$\rho_{\text{total}} = \rho_{\text{in}} + \delta\rho_{\text{defect}}, \qquad \rho_{\text{grav}} = \varepsilon\,\rho_{\text{in}} + \delta\rho_{\text{grav}}. \tag{7}$$

Only a fraction $\varepsilon \sim \rho_\Lambda/\rho_{\text{in}}$ of inertial density gravitates at cosmological scales, addressing the 10^{120} discrepancy **if** ρ_{in} is large and $\varepsilon \ll 1$. This is a mechanism sketch, not a fitted cosmology; ε is not determined by the lab protocol in this paper.

2.5 Gradient gate

$|\nabla\rho|$ is the magnitude of the spatial gradient of the vacuum density $\rho = |\psi|^2$ (Eq. 2): it measures how sharply ρ changes from place to place (large near boundaries and defects, ≈ 0 in uniform void). CH links Casimir ripple α_{eff}, interferometric dip ΔV, and dispersion β_{eff} to $|\nabla\rho|$ through the gate $\chi(|\nabla\rho|)$ (Eqs. 8 and 10).

The gate width w sets how sharply observables turn on in $\log_{10}(|\nabla\rho|/|\nabla\rho|_c)$; our analysis code uses $w \approx 0.35$ by default (Section 5.3) [7].

$$\chi(|\nabla\rho|) = \frac{1}{2}\left[1 + \tanh\left(\frac{\log_{10}(|\nabla\rho|/|\nabla\rho|_c)}{w}\right)\right], \tag{8}$$

$$|\nabla\rho|_c = \frac{\rho_{\text{in}}}{\xi} \quad \Rightarrow \quad |\nabla\rho|_c = \frac{c^2}{G} \approx 1.3\times 10^{27}\,\text{kg}\,\text{m}^{-4} \tag{9}$$

when $G = \alpha_G c_s^2 \xi/\rho_{\text{in}}$ with $\alpha_G = 1$, $c_s = c$. **Intuition:** $|\nabla\rho|_c$ is the density gradient at which the healing length ξ becomes comparable to the scale of variation in ρ—the natural boundary between "uniform fluid" and "structured" vacuum. Tidal gradients from a lead brick are $\sim 10^{-22}$ of this scale; Casimir walls at gap $d \sim \xi$ can approach $\mathcal{O}(1)$ (Section 5).

Gated observables:

$$\alpha_{\text{eff}} = \alpha_{\max}\,\chi, \qquad \Delta V = \Delta V_{\max}\,\chi, \qquad \beta_{\text{eff}} = \beta_{\max}\,\chi. \tag{10}$$

Dispersion ceiling (Prediction #4):

$$\beta_{\max} = \frac{3}{2}\,\frac{\xi^2}{\hbar^2}. \tag{11}$$

Here $\beta_{\max}$ is the maximum quadratic Lorentz-violation coefficient in the arrival-time law $\Delta t \approx (D/c^3)\,\beta_{\text{eff}}\,E^2$ (Section 4.2). For $\xi \sim 50$ nm, $\beta_{\max} \sim 10^{20}\,\text{s}\,\text{GeV}^{-2}$—orders of magnitude above Fermi bounds—so an **always-on** CH dispersion is excluded; gradient gating sends $\beta_{\text{eff}} = \beta_{\max}\chi \to 0$ on void paths.

3 From theory to bench observables

This section states the **measurement links** a skeptical experimentalist needs: how abstract GPE fields become numbers read off an AFM or interferometer.

3.1 Casimir ripple amplitude α (Prediction #3)

Standard Casimir theory gives a smooth $1/d^4$ force between parallel conductors [8]. CH **parameterizes** (does not derive from GPE) a gated multiplicative ripple on the normalized force:

$$\frac{F(d)}{F_{\text{Cas}}(d)} = 1 + \alpha_{\text{eff}}(k)\,\cos\left(\frac{2\pi d}{a_{\text{vac}}} + \phi\right), \tag{12}$$

where a_{vac} is a vacuum lattice scale (fit parameter) and $\alpha_{\text{eff}} = \alpha_{\max}\,\chi(|\nabla\rho|(k))$. **Extraction:** scan gap d, divide measured force by the QED Casimir prediction $F_{\text{Cas}}(d)$, fit the oscillatory residual for amplitude α (practice pipeline: `casimir_ripple_sim.py`). QFT predicts $\alpha_{\text{eff}} = 0$ within noise; CH predicts α_{eff} rises when the knob k raises $|\nabla\rho|$ past $|\nabla\rho|_c$.

3.2 Mach–Zehnder visibility dip ΔV (Prediction #6)

Standard environmental decoherence gives visibility $V(\Delta L) = e^{-\Gamma \Delta L}$ for path-length difference ΔL. CH adds a gated residual wobble after subtracting that envelope:

$$V_{\mathrm{meas}}(\Delta L) = e^{-\Gamma\Delta L}\left[1 + \varepsilon_{\mathrm{vac}}\,\chi\cos\left(\frac{2\pi\Delta L}{a_{\mathrm{vac}}}\right)\right], \qquad \Delta V(k) = \Delta V_{\max}\,\chi(|\nabla\rho|(k)). \tag{13}$$

Extraction: scan ΔL, fit $e^{-\Gamma\Delta L}$ to the envelope, define ΔV as the peak-to-peak residual visibility (practice: `mach_zehnder_visibility_sim.py`).

3.3 Photon dispersion β_{eff} (Prediction #4)

On a cosmological path of length D, a quadratic energy-dependent arrival-time offset is parameterized by

$$t(E) = t_0 + \frac{D}{c^3}\,\beta_{\mathrm{eff}}\,E^2, \tag{14}$$

with E in joules in the fit code ($E_{\mathrm{GeV}} \times 1.602 \times 10^{-10}$ J). Void paths have $\chi \approx 0$ so $\beta_{\mathrm{eff}} \approx 0$ even when $\beta_{\max}$ is huge (Eq. 11).

3.4 Knob k and observable $O(k)$ (Prediction #7)

Prediction #7 does **not** re-use the ripple fit of Eq. 12 directly; it compares the **scalar amplitudes** $\alpha(k)$ and $\Delta V(k)$ (already extracted per k) against a shared threshold in k-space:

$$O_{\mathrm{QFT}}(k) = c_0, \qquad O_{\mathrm{CH}}(k) = c_0 + A\,\mathcal{S}(k - k_c;\,\delta k), \tag{15}$$

where $\mathcal{S}(x) = \frac{1}{2}[1 + \tanh(x/\delta k)]$ is a smooth step and $\delta k \approx 0.08\,(k_{\max} - k_{\min})$ in the reference analysis. The critical knob value k_c is the laboratory proxy for the gradient gate; consistency requires the **same** k_c (within a factor of 2) for α and ΔV if both are measured.

3.5 Secondary predictions (#1, #2, #5)

For completeness: **#1** tests sidereal modulation of an effective speed of light (null expected in uniform ρ). **#2** tests whether Bell correlators vary with sidereal phase (optional; not central to CH). **#5** tests short-range Yukawa deviations from $1/r^2$ gravity inside cavities (localized gradients only). These are listed for completeness; the present protocol prioritizes #3, #6, and #7.

4 Predictions

Table 1: Seven CH predictions: uniform vs high-$|\nabla\rho|$ environments.

#	Observable	QFT (uniform)	CH (high $\lvert\nabla\rho\rvert$)	Priority
1	Sidereal c anisotropy	Null	Unlikely	Secondary
2	Bell S vs sidereal	Null	Optional	Secondary
3	Casimir F/F_{Cas} ripple	Null	$\alpha_{\mathrm{eff}} = \alpha_{\max}\chi$	Primary
4	GRB photon dispersion	Null	$\beta_{\mathrm{eff}} \approx 0$ (void)	Astro null
5	Yukawa fifth force	Null in cavity	Localized	Secondary
6	Fringe visibility wobble	Decoherence only	$\Delta V = \Delta V_{\max}\chi$	Primary
7	$O(k)$ vs gradient knob	**Flat**	**Threshold**	**Discriminating**

Predictions #1–#6 are often **null in quiet labs**; #7 asks whether they **turn on together** when a knob raises $|\nabla\rho|$. Table 1 summarizes all seven; Sections 3 and 6 define how #3, #6, and #7 are measured and compared.

4.1 GRB 090510: three dispersion models

Table 2: Void-path dispersion models vs Fermi LAT extended analysis of GRB 090510.

Model	Void-path prediction	GRB 090510 result
QFT + GR	$\beta = 0$	Consistent (null, $p \approx 0.56$)
Naive always-on CH	$\beta = \beta_{\max}$ (huge)	**Excluded**
Gradient-gated CH	$\beta_{\text{eff}} \approx 0$	Consistent (not proof)

Completed pipeline: 28 photons with $E \geq 1\,\text{GeV}$, $E_{\max} \approx 30\,\text{GeV}$, $\beta_{95} \approx 2 \times 10^{16}$ [9, 10, 7].

4.2 GRB analysis methods and interpretation

Data and cuts. We use Fermi **LAT extended** event data (TRANSIENT class) for trigger bn090510016 ($z = 0.903$), queried from the Fermi LAT Data Server [7]. Photons are selected in a $\pm 12°$ ROI, arrival times within 10 s of trigger, and $E \geq 1$ GeV, yielding $n = 28$ events ($E_{\max} \approx 29.9$ GeV). GBM-only ($n = 41$) and LAT LLE ($n = 19$) pipelines are also implemented for cross-checks; the extended LAT sample is closest to the high-energy literature on this burst [9, 10].

Fit model. Following the standard quadratic Lorentz-violation test [9], we linearly regress arrival time t against E^2 (Eq. 14) with $D = D_L(z = 0.903) \approx 5.99$ Gpc. The slope maps to $\beta = \text{slope} \times c^3/D$; we report the one-sided 95% upper limit $\beta_{95} = \beta + 1.96\,\sigma_\beta$ when β is consistent with zero.

Results. Best-fit β is consistent with zero ($p \approx 0.56$, reduced $\chi^2 \approx 5.7$). The elevated χ^2_{red} on $n = 28$ photons indicates the linear-in-E^2 template fits the burst poorly (intrinsic timing structure, not necessarily Lorentz violation); the null slope is therefore conservative with respect to CH but does not validate the functional form. $\beta_{95} \approx 2.1 \times 10^{16}\ \text{s}\,\text{GeV}^{-2}$ is far below naive $\beta_{\max}$ for any microscopic ξ, **excluding always-on CH dispersion**.

What this does not show. A null on a void path does **not** confirm CH—standard QFT also predicts $\beta = 0$. The analysis does **not** constrain ξ or ρ_{in} for gradient-gated CH, because $\chi \approx 0$ along the intergalactic path. The discriminating test remains Prediction #7 in the laboratory.

Always-on benchmark. If $\beta = \beta_{\max}$ everywhere with $\xi \lesssim 10^{-15}$ m, Fermi timing would show enormous dispersion; gradient gating is CH's mechanism to retain a large $\beta_{\max}$ while passing void-path tests.

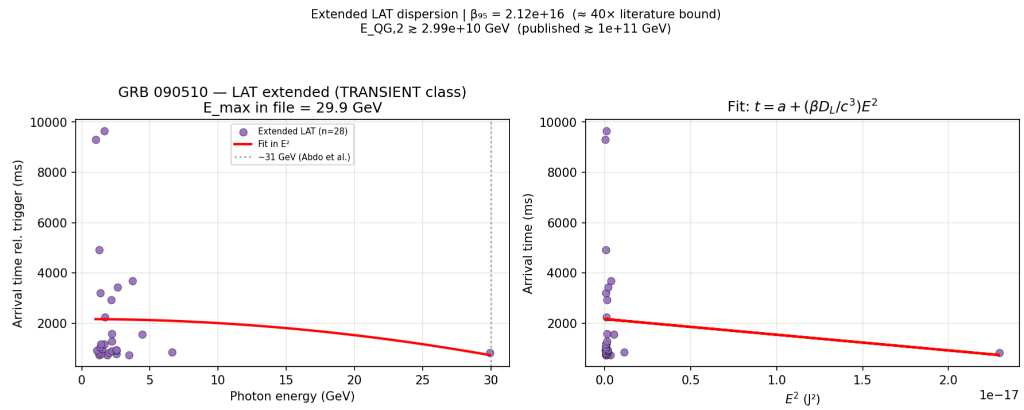

Figure 1: Real Fermi LAT extended analysis: photon arrival time vs E^2 for GRB 090510 ($z = 0.903$). Null fit ($p \approx 0.56$) is consistent with QFT and gradient-gated CH on void paths.

5 GPE boundary program

Tidal gradients from laboratory masses give $|\nabla\rho|/|\nabla\rho|_c \sim 10^{-22}$—unsuitable as an experimental knob. **Casimir boundaries** impose Dirichlet-like conditions on ψ at conductors and can raise local $|\nabla\rho|$ near walls when the gap d is comparable to the healing length ξ.

5.1 Thomas–Fermi boundary ansatz

Full numerical GPE solutions with defect sources are future work; for knob planning we use a **Thomas–Fermi (TF) boundary-layer ansatz** validated against 1D imaginary-time GPE in the companion code [7]. The ansatz enforces $\rho \to 0$ at walls and bulk $\rho \to \rho_{\text{in}}$ far from boundaries: **1D parallel-plate** ($\hat{d} = d/\xi$):

$$\rho(\hat{z}) \approx \tanh(\hat{z})\,\tanh(\hat{d} - \hat{z}), \qquad \frac{|\nabla\rho|}{|\nabla\rho|_c} = \left|\frac{\partial|\psi|^2}{\partial\hat{z}}\right|. \tag{16}$$

2D sphere–plate ($\hat{h}(\hat{r}) = \hat{d}_{\min} + \hat{r}^2/(2\hat{R})$):

$$\rho(\hat{r}, \hat{z}) \approx \tanh(\hat{z})\,\tanh(\hat{h}(\hat{r}) - \hat{z}). \tag{17}$$

At the sphere rim ($\hat{r} \sim 1$), radial gradients scale as $\sim 1/\hat{R}$—so **curvature radius** R is a second knob that modulates wall gradients without changing bulk gap midpoints in wide-gap limits.

Table 3: Illustrative GPE results ($\xi = 50\,\text{nm}$, wide gap $d \gg \xi$).

Location	Role in Prediction #7
Gap bulk (central half)	χ_{bulk} turns on when $d \lesssim \mathcal{O}(\xi)$ — **signal** when scanning d
Plate wall	χ_{wall} saturated, flat vs d — **control** on d scan; **signal** when scanning R
Tidal / lead brick	$\sim 10^{-22}$ of threshold — wrong knob

5.2 Probe coupling for gap vs curvature scans

GPE boundary forecasts attach χ to **distinct probe loci**. Prediction #7 uses them as follows (code flag `coupling=mid` maps bulk χ to both α and ΔV in gap demos; `coupling=wall` is the flat control forecast).

Gap scan (d knob, signal). Ripple amplitude $\alpha(d)$ from Eq. 12 is compared against $\chi_{\text{bulk}}(d)$: the gradient gate evaluated at the **maximum** $|\nabla\rho|$ in the central half of the Thomas–Fermi cavity profile. When $d \gg \xi$, wall layers decouple and $\chi_{\text{bulk}} \to 0$ (QFT-flat sector); when $d \lesssim \mathcal{O}(\xi)$, overlapping layers raise bulk gradients and χ_{bulk} turns on. The geometric gap midpoint is **not** used: for $\rho \propto \tanh(\hat{z})\tanh(\hat{d}-\hat{z})$ it sits at a symmetry zero when layers overlap.

Gap scan (control). $\chi_{\text{wall}}(d)$ is near saturation and flat vs d for $d \gtrsim 2\xi$; register α vs T or duplicate geometries as independent flat controls (Table 4).

Curvature scan (R knob, signal). At fixed minimum gap, scan sphere radius R and extract α at the **wall/rim** where GPE predicts χ_{wall} varies with R while bulk midpoints stay in the null sector (Eq. 17).

5.3 Forecast and analysis defaults

CH does not fix ξ, $\alpha_{\max}$, or per-shot ripple uncertainty from first principles. For laptop forecasts, figures, and the power study we adopt the following **planning defaults** (replaceable once collaborators register pilot F/F_{Cas} fits):

Healing length. $\xi = 50\,\text{nm}$ is a representative value in the hypothesized lab band $\mathcal{O}(10)$–100 nm (symbol table, Appendix A). Measured k_c or a_{vac} from Eqs. 12 would supersede it

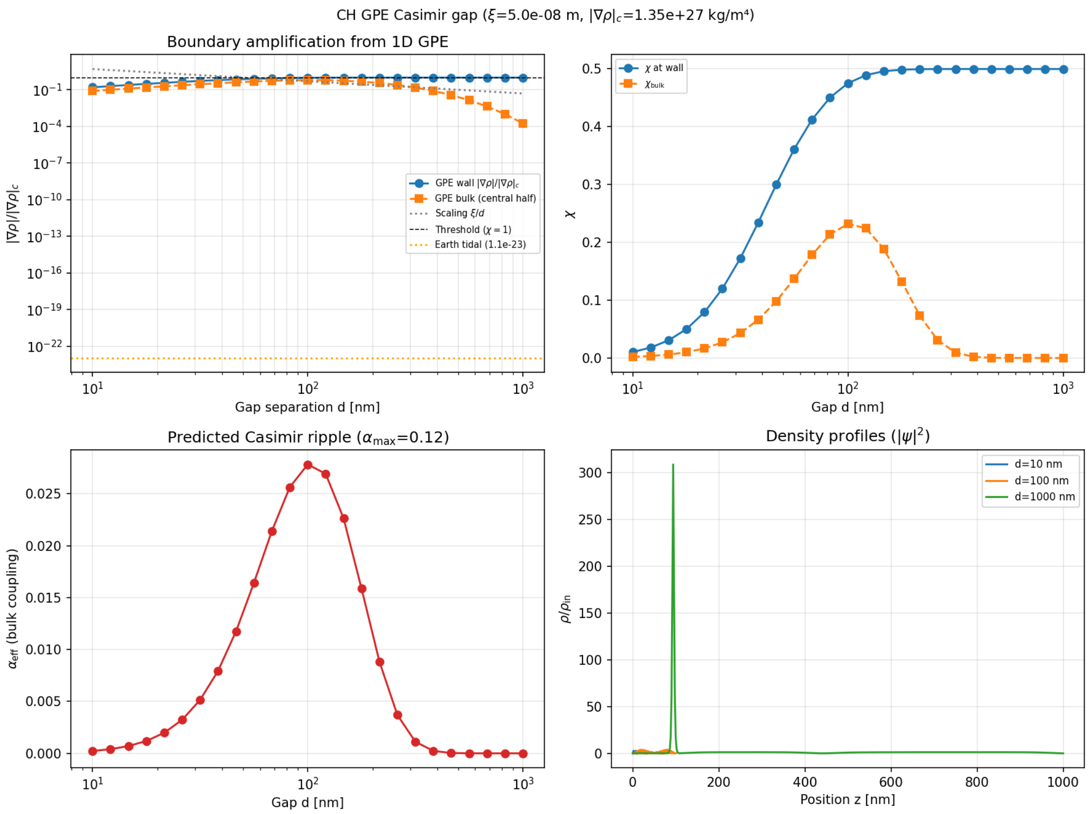

Figure 2: 1D GPE Casimir-gap scan ($\xi = 50\,\text{nm}$): wall vs bulk gradient ratio and χ vs gap d. Bulk χ uses the central-half maximum (Section 5.2), not the geometric midpoint alone.

via Section 6.7. The illustrative ripple period $a_{\rm vac} = 150$ nm in companion demos implies $\xi \approx a_{\rm vac}/(2\pi) \approx 24$ nm—the same order of magnitude, not assumed equal.

Gap scan endpoints (d = 40–600 nm). The lower bound matches the minimum controllable sphere–plate separations in dynamic Casimir AFM work; the upper bound follows the ripple-analysis convention in `casimir_ripple_sim.py` (50–600 nm scan, ~ 3.7 periods at $a_{\rm vac} = 150$ nm). At $\xi = 50$ nm this span gives $\hat{d} = d/\xi \approx 0.8$–12, placing gradient-gate turn-on inside the protocol range rather than below the minimum gap or in the asymptotically flat sector.

Ripple ceiling $\alpha_{\max} = 0.12$. Eq. 10 uses $\alpha_{\rm eff} = \alpha_{\max}\chi$; $\alpha_{\max}$ is an **illustrative coupling cap**, not a GPE output. The value 0.12 (12% on $F/F_{\rm Cas}$) is chosen in `casimir_ripple_sim.py` as "visible but still subtle" for practice fits—order-of-magnitude headroom for a gated ripple, not a derived prediction. The lab measures α directly; GPE supplies the shape $\chi(d)$, not the absolute scale.

Per-shot uncertainty $\sigma_\alpha = 0.01$. The power study (Section 8.9) adopts $\sigma_\alpha = 0.01$ on extracted ripple amplitude as a **moderate** planning assumption (companion checklist: optimistic 0.005, conservative 0.02). Twenty repeats per gap yield $\sigma_{\rm eff} \approx 0.0022$; collaborators should register σ_α from pilot data and re-run `ch_threshold_power_study.py` before unblinding.

Gate width $w \approx 0.35$. In Eq. 8, w is the transition width in $\log_{10}(|\nabla\rho|/|\nabla\rho|_c)$ for the tanh gate. The default 0.35 gives an $\mathcal{O}(1)$-decade turn-on in $\log_{10}$ gradient ratio and matches the smooth-step width used in threshold fits (Eq. 15); it is a numerical/analysis choice, not fitted from data in this paper.

5.4 GPE forecasts linked to Prediction #7

With $\xi = 50$ nm and gap scan $d = 40$–600 nm ($k = 10^9/d_{[\rm m]}$), the GPE pipeline (`ch_gpe_to_threshold_demo.py`) maps $d \to \chi_{\rm bulk}$ and $\chi_{\rm wall}$ and generates synthetic (k, α, σ) and $(k, \Delta V, \sigma)$ CSVs. Flat vs threshold fits (`gradient_threshold_analysis.py`) on that synthetic data yield $\Delta\chi^2 \approx 274$ (α) and $\Delta\chi^2 \approx 429$ (ΔV)—well above the detection cut $\Delta\chi^2 = 9$. This demonstrates that **if** CH gated amplitudes track GPE χ vs d under the bulk coupling of Section 5.2, the protocol has statistical power; it is **not** a substitute for real lab data.

Curvature scans (`ch_gpe_sphere_to_threshold_demo.py`) produce smaller χ variations but detectable threshold trends when R is scanned at fixed minimum gap; there the signal channel is **wall/rim-coupled** $\alpha(R)$ (Section 5.2).

6 Experimental protocol (Prediction #7)

6.1 Scientific question

As knob k scans (gap d or radius R), does observable $O(k)$ stay **flat** (QFT) or **turn on** above a threshold (gradient-gated CH)? This is a **model comparison**, not a single null test: both models may fit poorly if systematics dominate.

6.2 Measurement channels

Confirm CH (eventually): threshold in ≥ 1 signal channel, **same** k_c (within factor 2) across signal channels if both are measured, and **flat control** ($\Delta\chi^2 < 9$ for threshold vs flat on control).

Rule out CH (in scanned range): GPE forecasts turn-on over your k range but data prefer flat $O(k)$, **or** signal channels disagree on k_c by more than factor 2.

Table 4: Signal vs control channels for Prediction #7.

Channel	Type	How O is obtained
$\alpha(k)$	Signal (gap d)	Ripple amplitude from $F(d)/F_{\mathrm{Cas}}(d)$; GPE forecast uses $\chi_{\mathrm{bulk}}(d)$ (Section 5.2)
$\alpha(R)$	Signal (curvature)	Wall/rim-coupled ripple when scanning R at fixed minimum gap
$\Delta V(k)$	Signal	Peak residual visibility after $e^{-\Gamma \Delta L}$ subtraction (Eq. 13)
α vs T	Control	Same geometry; T is not a CH gradient knob
χ_{wall} vs d	Control	GPE predicts flat $\chi_{\mathrm{wall}}(d)$ when scanning gap
α vs lateral offset	Control	When GPE predicts unchanged $\lvert\nabla\rho\rvert$

Table 5: Laboratory knobs and measurement channels.

Knob	Definition	Example range	Role
Gap d	$k = 10^9/d_{[\mathrm{m}]} = 1/d_{\mathrm{nm}}$	40–600 nm	Signal
Curvature R	$k = 10^6/R_{[\mathrm{m}]} = 1/R_{\mu\mathrm{m}}$	0.05–5 µm	Signal
Temperature T	same k grid as signal run	—	Control

6.3 Knob definitions

Higher k means smaller d or smaller R—i.e. stronger boundary confinement and (in GPE forecasts) larger $\lvert\nabla\rho\rvert$ near walls.

6.4 Statistical models and detection cuts

Each channel carries tuples (k_i, O_i, σ_i). We fit Eqs. 15 with weighted χ^2:

$$\chi^2_{\mathrm{flat}} = \sum_i \frac{[O_i - c_0]^2}{\sigma_i^2}, \tag{18}$$

$$\chi^2_{\mathrm{thr}} = \sum_i \frac{[O_i - O_{\mathrm{CH}}(k_i; c_0, A, k_c)]^2}{\sigma_i^2}. \tag{19}$$

Detection: $\Delta\chi^2 = \chi^2_{\mathrm{flat}} - \chi^2_{\mathrm{thr}} \geq 9$ (two extra parameters: amplitude A and k_c; $\approx 3\sigma$ rule of thumb). **Joint fit:** when both α and ΔV are available, refit with a **shared** k_c and separate amplitudes per channel (`gradient_threshold_analysis.py --joint`).

Pre-registered requirements (complete before data collection):

- ≥ 10 distinct k settings spanning the GPE-predicted turn-on range.
- ≥ 20 independent repeats per $(k, \text{channel})$ when possible.
- Control channel chosen and registered **before** unblinding signal fits.
- $\Delta\chi^2$ cut $= 9$; k_c agreement factor $= 2$; control pass: $\Delta\chi^2 < 9$.

Full printable checklist: companion document `ch-lab-protocol-checklist.md` [7].

6.5 Verdict logic

6.6 Hardware tiers

- **Tier A:** Dynamic AFM Casimir — α only (minimum viable; collaboration).

Table 6: Protocol outcomes (implemented in `control_channel_analysis.py`).

Outcome	Interpretation
Signal threshold + flat control + same k_c	Consistent with gradient-gated CH
All flat over GPE turn-on range	CH gradient sector ruled out (in range)
Different k_c in α vs ΔV	Single $\lvert\nabla\rho\rvert_c$ excluded
Control thresholds like signal	Likely systematic

- **Tier B:** Tier A + matter-wave MZ — $\alpha + \Delta V$.
- **Tier C:** Tunable d, R, and control runs — full #7 protocol.

Data format: CSV with header `k,O,sigma` (one row per k setting after averaging repeats).
Analysis command (replace paths with lab exports):

```
python3 control_channel_analysis.py \
  --csv-alpha lab/run001_alpha.csv \
  --csv-vis   lab/run001_vis.csv \
  --csv-control lab/run001_control.csv
```

Demo with synthetic data: `control_channel_analysis.py --demo`. On demo data the script reports $\Delta\chi^2 \approx 192$ (α), $\Delta\chi^2 \approx 244$ (ΔV), control $\Delta\chi^2 \approx 1.3$, joint $\Delta\chi^2 \approx 436$, verdict `CONSISTENT WITH GRADIENT-GATED CH` [7].

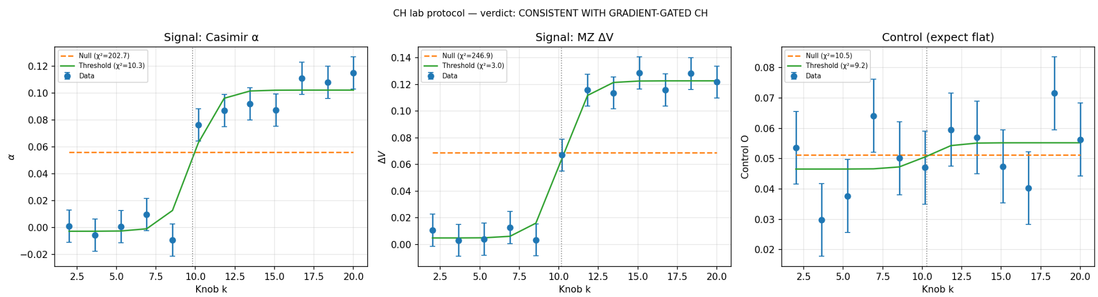

Figure 3: Example protocol analysis: signal channels (threshold fit) and control channel (flat). Demo data from `control_channel_analysis.py --demo`.

6.7 From protocol k_c to ξ and gravity forecasts

Prediction #7 constrains the **turn-on knob** k_c, not the healing length ξ directly. CH does not yet predict ξ uniquely from first principles; instead we implement **identification hypotheses** that map laboratory outputs to ξ and then to emergent-gravity diagnostics [7]. The value $\xi = 50\,\mathrm{nm}$ used in GPE figures and power forecasts is a planning default only (Section 5.3).

Step 1 — extract k_c. After threshold detection, the joint fit (`gradient_threshold_analysis.py --joint`) returns a shared k_c with per-channel amplitudes. For gap scans, $k = 10^9/d_{[\mathrm{m}]} = 1/d_{\mathrm{nm}}$ (Table 5), so the fitted k_c implies a turn-on gap

$$d_c \approx \frac{1}{k_c}\,\mathrm{nm} = \frac{1}{k_c \times 10^9}\,\mathrm{m}. \tag{20}$$

Step 2 — ξ hypotheses. Two mechanisms are cross-checked in `ch_xi_prediction.py`:

- **Gap turn-on:** $\xi \approx d_c$ where the gradient gate turns on in the scanned knob (Prediction #7 $\leftrightarrow$ GPE wall saturation).

- **Casimir lattice:** $\xi = a_{\text{vac}}/(2\pi)$ from the ripple period a_{vac} in Eq. 12 (Prediction #3).

If both are available and agree within a factor of 2, the pipeline reports their geometric mean; otherwise the gap-turn-on value from k_c is used. Each hypothesis fixes (ξ, ρ_{in}) via Eq. 6 with $\alpha_G = 1$ and yields a consistency report ($\beta_{\max}$, always-on Fermi exclusion, gated void PASS).

Step 3 — gravity forecast with α_G calibration. Spherical defect solves (`ch_gpe_gravity.py`, solver `v3_sm_matched`) use a matter sink S_M on an inner domain and a Schwarzschild-matched boundary at r_{join}—no post-hoc density tail patch. Exterior Madelung Q contributes negligibly; the hydrostatic $\delta\rho$ channel dominates and, at $\alpha_G = 1$, overshoots Newton's law by many orders of magnitude (the same tension noted in Section 8). Given a fixed density profile, we therefore **solve for** α_G such that the total acceleration satisfies $|a|/(GM/r^2) \approx 1$ in the fit annulus:

$$\alpha_G^{\text{cal}} \approx \frac{1 - F_Q}{F_{\text{hydro}}(\alpha_G = 1)}, \tag{21}$$

where F_Q and F_{hydro} are the fitted Newton factors from the quantum and hydrostatic channels. This is a **matching calibration**, not a first-principles derivation of α_G; it shows how a single lab-fixed ξ propagates to a self-consistent far-field check once the defect profile is known.

End-to-end pipeline. `ch_lab_pipeline_demo.py` runs protocol verdict $\to k_c \to \xi \to$ gravity v3 $\to \alpha_G^{\text{cal}}$ on demo CSVs. On synthetic demo data ($k_c \approx 10$ nm^{-1}, joint $\Delta\chi^2 \approx 436$), gap turn-on gives $\xi \approx 0.1$ nm and calibrated Newton factor ≈ 0.91 [7]. Casimir-lattice ξ from $a_{\text{vac}} = 150$ nm ($\xi \approx 24$ nm) disagrees with this synthetic k_c by design—illustrating that the two hypotheses must be tested against real data, not assumed identical.

7 Software and reproducibility

All analyses cited in this paper are implemented in the open GitHub repository https://github.com/G-Web-Essentials/CH-A-Gross-Pitaevskii-Superfluid-Vacuum [7]. **Recommended reproduction order** for a skeptical reader:

1. `fermi_grb090510_lat_extended_beta_limit.py` — download/query LAT extended events, fit β, reproduce Fig. 1.
2. `ch_gpe_casimir_gap.py` — wall vs bulk χ vs d at $\xi = 50$ nm (Fig. 2).
3. `ch_gpe_to_threshold_demo.py` — GPE $\to$ CSV $\to \Delta\chi^2$ for gap knob.
4. `control_channel_analysis.py --demo` — full protocol verdict (Fig. 3).
5. `ch_lab_pipeline_demo.py` — $k_c \to \xi \to$ gravity v3 + α_G calibration (Section 6.7).
6. `ch_threshold_power_study.py` — Monte Carlo 90% power vs $\alpha_{\max}$, n_k, σ_α (Section 8.9).

Key scripts:

- `ch_dispersion_fermi_combined.py` — real GRB 090510 + $\beta(\xi)$ overlay;
- `ch_gpe_to_threshold_demo.py` — GPE $\to$ CSV $\to$ fit (gap d);
- `ch_gpe_sphere_to_threshold_demo.py` — curvature R pipeline;
- `gradient_threshold_analysis.py` — flat vs threshold fits;
- `control_channel_analysis.py` — signal + control verdict;
- `ch_xi_lab_bridge.py` — protocol CSVs $\to \xi$ hypotheses $\to$ gravity forecast;

- `ch_gpe_gravity_demo.py` — spherical defect v3 + α_G calibration;
- `ch_xi_prediction_demo.py` — compare ξ identification mechanisms;
- `ch_threshold_power_study.py` — Monte Carlo detection power vs $\alpha_{\max}$, n_k, σ;
- `casimir_ripple_sim.py` — practice ripple extraction.

8 Discussion

8.1 Null results and quiet vacuum

Eöt-Wash nulls [11], smooth Casimir forces [8], and GRB timing nulls [12, 9] are **expected** in uniform ρ. CH is challenged by **gradient-correlated** positives, not by another void null.

8.2 Always-on vs gradient-gated

Naive $\beta = \beta_{\max}$ everywhere is **excluded** by Fermi. Gradient gating sets $\chi \to 0$ on void paths and $\chi \to 1$ only where $|\nabla\rho|$ is large [3, 1].

8.3 What empirical success would and would not imply

If Prediction #7 were confirmed—threshold turn-on in $\alpha(k)$ and/or $\Delta V(k)$, flat control, and consistent k_c across signal channels—the established claim would be **narrow but profound**: the vacuum exhibits a gradient-gated sector (supersolid-linked observables that QFT does not predict) turning on when boundary physics raises $|\nabla\rho|$ past a single critical scale. Shared k_c and agreement with GPE boundary forecasts would support one healing length ξ linking lab turn-on, Casimir ripple period, and emergent-gravity matching sketches (Section 6.7).

That result would **not** by itself validate the full ontological program of Section 1: space as a material BEC, matter as defects in ρ, emergent G and clocks, or the two-component cosmological-constant mechanism. Those interpretations organize null results in uniform vacuum and motivate the lab knob class, but they remain extensions requiring independent theoretical and empirical work (fermions, unique ξ, full defect gravity). In short: a positive #7 would evidence **gated vacuum structure**, not a complete replacement of QFT and general relativity.

8.4 Roadmap beyond Prediction #7

Readers new to this program may wonder: *if the threshold test succeeds, what comes next?* CH is intentionally staged. Prediction #7 (Table 1) is the **wedge**—it asks whether any vacuum observable turns on when a laboratory knob raises $|\nabla\rho|$. The broader framework in Section 1—material vacuum, defect matter, emergent G and time, cosmological ε—requires **additional** tests and theory at each layer. None of these layers is automatic even after a positive #7; each must earn its evidence.

Layer 1: Gradient-gated vacuum sector (near-term lab). A confirmed #7 would show that at least one observable (α or ΔV) follows a threshold in knob k, with a flat control channel ruling out simple systematics. To strengthen the case that the vacuum is **structured** with a single length scale ξ, rather than an arbitrary fit:

- **Prediction #3 + #7:** Casimir ripple period a_{vac} (Eq. 12) and threshold k_c should map to the **same** ξ via the identification hypotheses of Section 6.7 ($\xi \approx d_c$, $\xi = a_{\text{vac}}/(2\pi)$).
- **Prediction #6 + #7 (Tier B/C):** Mach–Zehnder visibility dip $\Delta V(k)$ must turn on at the **same** k_c as $\alpha(k)$ within the pre-registered factor of 2—ruling out channel-specific artifacts.

- **Curvature knob R:** sphere–plate scans (`ch_gpe_sphere_to_threshold_demo.py`; full GP cross-check `run_sphere_plate_full_gp_scan.m`) should reproduce GPE-predicted trends without refitting ξ per geometry; laptop overlay shows radial rim agreement with Thomas–Fermi to $\sim$20% (Section 8.5).

- **Prediction #5:** short-range Yukawa or fifth-force searches **inside** high-χ cavities only; null results elsewhere are expected in CH and help separate gated from always-on sectors.

Outcome if these pass: a **gradient-gated supersolid-linked sector** with one healing length ξ is empirically established in boundary-stressed vacuum. This is still not proof that space *is* a BEC—only that it *behaves* as if a gated condensate order parameter is active where gradients are large.

Layer 2: Emergent Newton constant G (gravitation + numerics). CH relates $G = \alpha_G c_s^2 \xi / \rho_{\text{in}}$ (Eq. 6). Layer 1 would fix ξ in SI units; ρ_{in} and α_G then become **predictions**, not free parameters. Further work:

- **Defect gravity:** full three-dimensional GPE with matter source $S_M(\mathbf{x})$ (beyond the spherical v3 matched-BC sketch) must yield $|a| \approx GM/r^2$ from derived α_G, not post-hoc calibration alone (Section 6.7).

- **Mass–deficit check:** integrated density depletion $(1 - |\psi|^2)$ over a defect profile should match gravitational mass inferred from torsion-balance or free-fall experiments.

- **Equivalence principle:** all compositions should couple to the gated sector and to far-field gravity through the same ρ-defect picture; precision Eöt-Wash–type tests constrain species-dependent couplings.

- **Laptop consistency (done, not proof):** `ch_gpe_bh_exterior_demo.py` plots ρ, Φ, g_{eff} through r_s on the v3 matched exterior; see Section 8.5.

Casimir ripple experiments alone cannot measure G; this layer requires **collaboration with gravitation metrology** tied to the same ξ from Layer 1.

Layer 3: Emergent time and dispersion (clocks and high-energy lab). In CH, clocks track local condensate phase evolution (the "Chronos" in the name), and $\beta_{\text{eff}} = \beta_{\max}\chi$ (Eq. 11) gates Lorentz-violating dispersion. Void-path GRB nulls (Section 4.2) already constrain always-on dispersion; Layer 3 asks for **positive, gradient-correlated** tests in the laboratory:

- **Lab dispersion:** energy-dependent photon or particle arrival times in cavities where $\chi(k)$ is raised by the same knob as α and ΔV—a local test of Prediction #4, not another intergalactic null.

- **Clocks vs χ:** optical or matter-wave clock shifts correlated with k and with $\alpha(k)$, $\Delta V(k)$ at the same k_c.

- **Acoustic metric:** low-energy excitations propagating as if on the fluid metric $g_{\mu\nu}(\rho, c_s, \mathbf{v})$ in stressed regions (analog-gravity program [1, 2]).

- **Laptop consistency (done, not proof):** `ch_clock_redshift_from_gpe.py` and `ch_acoustic_light_bendi` show metric/Φ lensing and clock proxies are qualitatively consistent on matched defect exteriors but **exclude** naive $n \propto 1/\sqrt{\rho}$ lensing; see Section 8.5 and companion `universal-laws.tex`.

These are longer-horizon than Tier A Casimir but are the path from "gated ripple" to "time is hydrodynamic."

Layer 4: Defect ontology (matter as depletion in ρ). The postulate that electrons and nuclei are **stable defect patterns** in ψ, not objects in passive space, is the deepest ontological claim and the least directly accessible to Casimir hardware. Indirect support would require:

- composition-independent coupling to the gated sector and to emergent gravity (Layer 2);
- GPE defect modes predicting mass scales tied to ξ without per-species fitting;
- eventual theoretical embedding of fermions and gauge fields in ψ (Section 8.6 below).

A positive Layer 1–2 program would make defect ontology **plausible**; proving it is a separate, multi-decade theory-and-experiment effort.

Layer 5: Cosmological ε (two-component gravity). The split $\rho_{\mathrm{grav}} = \varepsilon\rho_{\mathrm{in}}$ (Eq. 7) addresses why enormous vacuum energy does not all curve spacetime: only $\varepsilon \ll 1$ gravitates cosmologically. This is **not** testable in a Casimir cell. It requires cosmological data—expansion history, CMB, structure formation—with ρ_{in} and ε constrained by Layer 1–2 inputs. CH would predict an order of magnitude for $\varepsilon \sim \rho_\Lambda/\rho_{\mathrm{in}}$ once ρ_{in} is fixed; falsification would come from failed cosmological fits, not from Prediction #7 alone.

Consistency checks (secondary predictions). Predictions #1–#2 (sidereal anisotropy in c or Bell correlators) assume a **global** lattice visible everywhere. CH's gradient-gated design expects these to remain **null** in quiet laboratories. If #7 passes while #1–#2 stay null, that supports local gating over always-on global order; a positive sidereal signal would require reconciling with the gated picture.

Table 7: Roadmap from Prediction #7 toward the full CH framework (for readers joining the program mid-stream).

Layer	Goal	Principal further tests
0 (this paper)	Threshold vs flat	#7: $\alpha(k)$, $\Delta V(k)$, control, joint k_c
1	Gated sector + ξ	#3+#6+#7 same k_c; R scan; #5 in cavity
2	Emergent G	Defect GPE $\to$ Newton; mass deficit; EP tests
3	Emergent time	Lab $\beta_{\mathrm{eff}}(k)$; clocks vs χ; acoustic metric
4	Defect matter	Composition independence; defect spectrum; SM embedding
5	Cosmological ε	CMB, $H(z)$, ε from ρ_{in}

Practical sequencing. We recommend: Tier A #7 $\to$ Tier B add ΔV $\to$ lock ξ from ripple + turn-on $\to$ gravitation and dispersion programs in parallel $\to$ cosmology only after ρ_{in} is constrained. Each stage publishes a **narrower** claim than the full ontology; that discipline keeps the program falsifiable and legible to collaborators who were not involved in its development.

8.5 Laptop consistency checks (Layers 2–3, not empirical proof)

Companion document `universal-laws.tex` summarizes how familiar laws are reinterpreted in CH. The scripts below test **internal consistency** of matched defect exteriors and acoustic-metric sketches—they do **not** validate CH empirically and do **not** substitute for Prediction #7.
Supported (qualitatively): defect Newton matching; attractive lensing via $n(\Phi)$ or $n \approx 1-r_s/r$; large-n Kepler correspondence; **sphere–plate curvature:** axisymmetric GP (`slice1d` mode, $\hat{R} \gtrsim 3$) matches Thomas–Fermi on the radial rim probe to median ∼18%, max ∼23% over $R \in [1, 100]\,\mu$m at $d_{\min} = 100$ nm, $\xi = 50$ nm (overlay `ch_gpe_sphere_plate_full_gp_overlay.py`).
Ruled out (naive readings): refractive index $n \propto 1/\sqrt{\rho}$ for gravitational lensing; a single ρ-to-observable map for both clocks and light; literal low-n planetary vortex radii at lab ξ; **laptop full-GP wall $|\nabla\rho|$ at $\hat{d}_{\min} \sim 2$** on coarse z grids (boundary layers unresolved—Casimir gap forecasts use TF + fine ξ-grid in Python).
Still open: full GR from GPE without matched exterior; clocks vs χ in cavity; BH interior; all gated positives at k_c; coupled 2D GP at tight curvature ($\hat{R} \lesssim 3$).

Table 8: Laptop numerics on defect exteriors (June 2026 repo). Outcomes are qualitative consistency checks, not laboratory evidence.

Script	Layer	Headline outcome
`ch_gpe_bh_exterior_demo.py`	2	ρ, Φ, g_{eff} vs r/r_s on v3 matched BC
`ch_clock_redshift_from_gpe.py`	3	$\sqrt{\rho}$, Φ clock proxies on analytic tail; lab $\chi(k)$ untested
`ch_acoustic_light_bending.py`	3	Metric/Φ index attracts ($\sim 1\times$ weak-field ref.); $n \propto 1/\sqrt{\rho}$ repels
`ch_vortex_kepler_toy.py`	4 sketch	$n \sim 10^{14} \to$ smooth Kepler orbits; low-$n \neq$ planets
`run_sphere_plate_full_gp_scan.m`	#7 / Tier-C	Full GP vs TF: radial rim $\lvert\partial\rho/\partial r\rvert$ within $\sim 20\%$ for R scan ($\hat{d}_{\min} = 2$); wall probe not validated on laptop grid

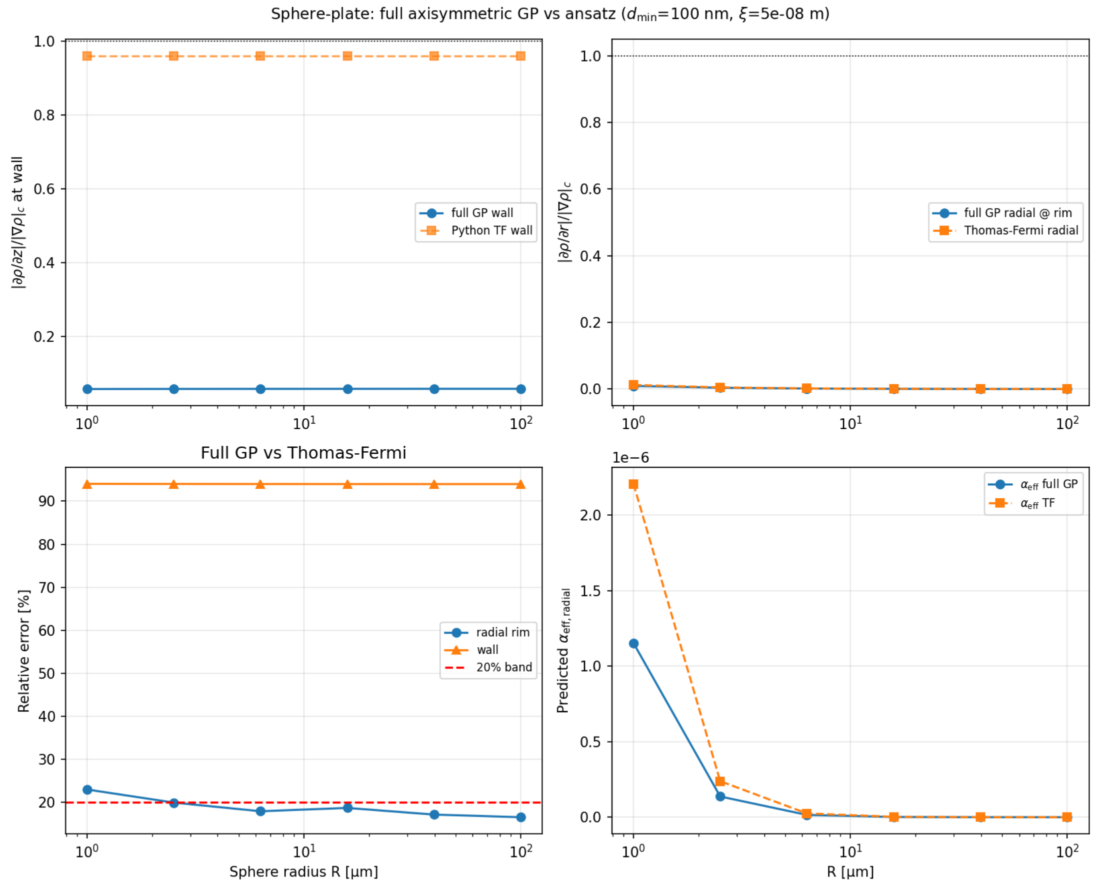

Figure 4: Sphere–plate full GP vs Thomas–Fermi (MATLAB `run_sphere_plate_full_gp_scan.m`, overlay `ch_gpe_sphere_plate_full_gp_overlay.py`; $d_{\min} = 100$ nm, $\xi = 50$ nm, $\hat{d}_{\min} = 2$). Radial rim probe: median relative error $\sim 18\%$, max $\sim 23\%$; wall probe not validated on this grid. Source: `simulations/output/ch_sphere_plate_full_gp_overlay.png`.

8.6 Open problems

1. **First-principles EM–GPE derivation of the Casimir ripple** (Eq. 12 is parameterized in this paper; companion roadmap: `docs/ch-paper2-em-gpe-casimir-derivation.md`).

2. Full GPE solution with defect source $S_M(\mathbf{x})$ (v3 uses matched BC continuation; nonlinear 3D solve remains open).

3. Fermions and gauge fields from ψ.

4. Unique prediction of ξ from first principles (identification hypotheses in Section 6.7 are testable but not yet derived).

5. Quantitative map $k \to |\nabla\rho|$ for specific AFM geometries.

6. **First-principles derivation of metric/Φ lensing and clock maps from the GPE** (laptop scripts use matched exteriors and chosen $n(\Phi)$; naive $n \propto 1/\sqrt{\rho}$ is excluded—Section 8.5; sphere–plate full GP vs TF on radial rim probe is done, wall probe at $\hat{d}_{\min} \sim 2$ is not).

8.7 Independent researchers

Theory, GPE forecasts, protocol, and analysis can be developed independently; Casimir hardware requires collaboration. A practical offer: pre-registered analysis and co-authorship in exchange for $F(d)$ data and metadata.

8.8 When laptop work warrants a lab attempt

Laptop simulations do **not** validate CH; they show the discriminating test is well posed and not already excluded in the wrong experimental regime. Four independent reasons justify a **narrow, pre-registered** Tier A attempt (dynamic AFM Casimir, α channel only):

Table 9: Why the gradient-threshold program warrants a targeted lab collaboration (not a major facility build).

Lens	Rationale
Scientific	CH predicts threshold turn-on correlated across signal channels when the vacuum density gradient exceeds $|\nabla\rho|_c$; QFT predicts flat $O(k)$. This is a direct model comparison, not another void-path null.
Methodological	Knob definitions, flat vs threshold fits, $\Delta\chi^2 \geq 9$ cut, k_c agreement (factor 2), and mandatory flat control are pre-registerable. Outcomes rule CH in **or** out in the scanned range.
Physical	GPE boundary work: tidal gradients $\sim 10^{-22}$ of threshold; Casimir walls at $d \sim \xi$ reach $\chi_{\text{wall}} \sim \frac{1}{2}$. If gradient gating is real, gap d (or curvature R) is the right knob class.
Practical	Tier A ($\alpha(k)$ only) is minimum viable; full Mach–Zehnder interferometry is not required to start. Collaborators export `k,O,sigma` CSVs; we return a verdict.

What is not established: no positive CH signal in real data; ξ not fixed from first principles; geometry-specific $k \to |\nabla\rho|$ maps remain open; gravity matching is calibration, not derivation; laptop Layer 2–3 checks (Table 8) are consistency sketches only.

8.9 Sensitivity and required scan density

To translate "we should try" into "we need N shots at these gaps," `ch_threshold_power_study.py` runs Monte Carlo trials on GPE-predicted $\alpha_{\text{eff}}(k) = \alpha_{\max}\chi_{\text{bulk}}(d)$ with realistic ripple uncertainty σ_α per shot, n_{repeat} averages per k, and n_k gap settings. Detection is declared when

$\Delta\chi^2 \geq 9$ (same cut as the protocol). Default scan and coupling assumptions (ξ, $\alpha_{\max}$, σ_α, gap range) are defined in Section 5.3. **Bulk** χ coupling maps $\alpha(d)$ to $\chi_{\text{bulk}}(d)$; $\chi_{\text{wall}}(d)$ is flat vs d and serves as a control forecast (Section 5.2).

Table 10 lists results from a full run ($n_{\text{trials}} = 400$). At $\alpha_{\max} = 0.12$ the detection power is 100% with median $\Delta\chi^2 \approx 191$. The bracketed minimum $\alpha_{\max}$ for 90% power is 0.037 ($\approx 3.7\%$ ripple ceiling) at $\sigma_\alpha = 0.01$; conservative $\sigma_\alpha = 0.02$ raises the ceiling to 0.074. Power remains $\geq 90\%$ for $n_k = 6$–20 at $\alpha_{\max} = 0.12$—the limiting factor at moderate noise is $\alpha_{\max}$, not scan density, once ≥ 6 gap settings span the GPE turn-on.

Table 10: Monte Carlo threshold-detection power (`ch_threshold_power_study.py`, $\xi = 50\,\text{nm}$, $d = 40$–$600\,\text{nm}$,n$_k = 12$, $n_{\text{repeat}} = 20$, $n_{\text{trials}} = 400$, bulk coupling).

Quantity	Result
σ_α per shot	0.010 (effective 0.0022 after 20 repeats)
Power at $\alpha_{\max} = 0.12$	100%; median $\Delta\chi^2 \approx 191$
Min. $\alpha_{\max}$ for 90% power ($\sigma_\alpha = 0.01$)	0.037
Min. $\alpha_{\max}$ for 90% power ($\sigma_\alpha = 0.02$)	0.074
Power vs n_k at $\alpha_{\max} = 0.12$	100% for $n_k \in \{6, 8, 10, 12, 14, 16, 20\}$

Table 11: GPE-recommended gap scan for Tier A ($\alpha_{\max} = 0.12$, bulk coupling; subset of 12 log-spaced settings). α_{eff} peaks near $d \sim 100\,\text{nm}$ and falls monotonically at larger d (no spurious high-d turn-on).

d [nm]	$k = 1/d_{\text{nm}}$ [nm^{-1}]	α_{eff}
40	0.0250	0.0087
84	0.0119	0.026
107	0.0093	0.028
137	0.0073	0.025
224	0.0045	0.0076
367	0.0027	~ 0
600	0.0017	~ 0

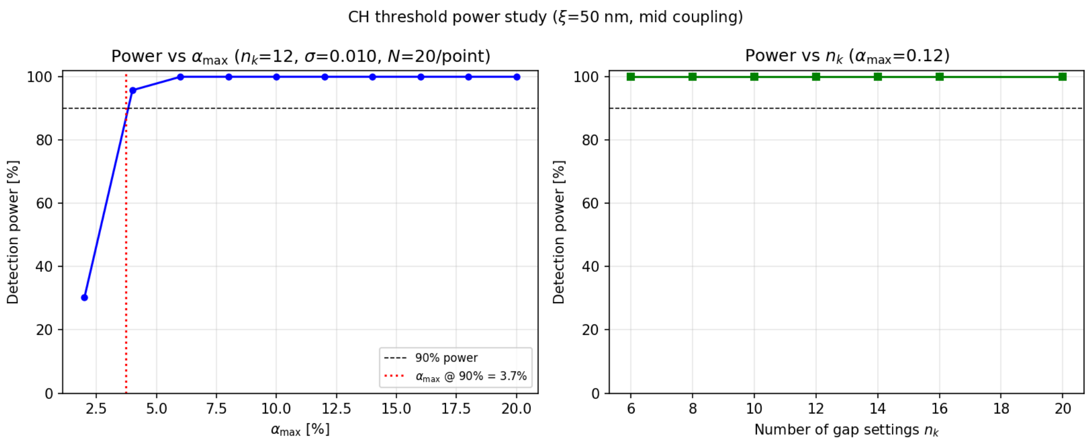

Figure 5: Monte Carlo detection power vs $\alpha_{\max}$ and vs number of gap settings n_k (`ch_threshold_power_study.py`). Dashed line: 90% power target.

Collaboration ask: register σ_α from pilot F/F_{Cas} fits, run the power study, then commit to $n_k \geq 10$ gaps and $n_{\text{repeat}} \geq 20$ per $(k, \text{channel})$ before unblinding threshold fits.

9 Conclusions

Chronos-Hydrodynamics proposes a physical BEC vacuum that is **invisible when uniform** and **testable when stressed**. Astrophysical nulls constrain always-on extensions; Prediction #7—threshold behavior in α and ΔV vs laboratory knobs—is the discriminating test. We supply GPE boundary estimates, a pre-registered protocol, real-data null analysis, and open verdict software. Confirmation requires collaborative Casimir (and ideally interferometric) data with a flat control channel and consistent k_c.

Acknowledgments

Computations used open-source Python (`numpy`, `matplotlib`) on a laptop workstation. Code and protocol scripts: https://github.com/G-Web-Essentials/CH-A-Gross-Pitaevskii-Superfluid-Vacu

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Symbol	Meaning
ψ	Vacuum order parameter; $\rho = \|\psi\|^2$
ρ_{in}	Inertial superfluid density (GPE bulk scale)
$\|\nabla\rho\|$	Magnitude of $\nabla\rho$; how sharply ρ varies in space (Eq. 2; gradient-gate subsection)
ξ	Healing length; hypothesized $\mathcal{O}(10)$–100 nm in lab forecasts
m_{grain}	$\hbar c_s/(c^2\xi)$; inertial mass of a vacuum grain
$\|\nabla\rho\|_c$	$\rho_{\text{in}}/\xi \approx c^2/G$; gradient gate scale
χ	Gradient gate $\in [0,1]$ (Eq. 8)
w	Gate width in $\log_{10}(\|\nabla\rho\|/\|\nabla\rho\|_c)$; default ≈ 0.35 (Section 5.3)
$\alpha_{\max}$, $\Delta V_{\max}$	Gated observable ceilings before χ; $\alpha_{\max} = 0.12$ is illustrative (Section 5.3)
α_{eff}, ΔV	$\alpha_{\max}\chi$, $\Delta V_{\max}\chi$
$\beta_{\max}$	$(3/2)\xi^2/\hbar^2$; dispersion ceiling
β_{eff}	$\beta_{\max}\chi$; used in GRB fits
k	Experimental knob; $10^9/d_{[\text{m}]}$ or $10^6/R_{[\text{m}]}$
k_c	Threshold knob value in Eq. 15
$O(k)$	Scalar observable at knob k (α or ΔV after extraction)
Γ	Environmental decoherence rate in MZ envelope
ε	Fraction of ρ_{in} that gravitates cosmologically

A Symbol table

B Fit models and verdict rules

Flat (QFT) model: $O(k) = c_0$.

Threshold (CH) model:

$$O(k) = c_0 + A \cdot \tfrac{1}{2}\left[1 + \tanh\left(\frac{k - k_c}{\delta k}\right)\right], \tag{22}$$

with $\delta k = \max[0.08\,(k_{\max} - k_{\min}),\ 10^{-12}]$.

Joint signal fit: one k_c, separate (c_0, A) per channel.

Verdict enum (`control_channel_analysis.py`):

- `CONSISTENT WITH GRADIENT-GATED CH` — signal threshold(s), flat control, k_c agree.
- `CH GRADIENT SECTOR RULED OUT (in scanned range)` — flat preferred where GPE predicted turn-on.
- `SINGLE` $|\nabla\rho|_c$ `EXCLUDED` — signal channels threshold but k_c disagree $> 2\times$.
- `LIKELY SYSTEMATIC` — control shows threshold like signal.
- `INCONCLUSIVE` — insufficient signal or failed control.

C Collaboration brief

Seeking: Existing or new $F(d)$ Casimir data, ≥ 10 gap settings, ripple amplitude α; optional temperature series as control.

Offering: Pre-registered analysis (`control_channel_analysis.py`), open repository, co-authorship.

Not required from lab: New theory development—only measurements and metadata.